

Conversational Image Recognition Chatbot

Team Lead: Utkarsh Kishor

1. Introduction & Problem Statement

In the rapidly evolving field of artificial intelligence, conversational agents (chatbots) have become ubiquitous. However, their capabilities are largely confined to the textual domain. A significant gap exists in their ability to comprehend and discuss visual information. The challenge is to bridge the gap between natural language processing (NLP) and computer vision, creating a system that can "see" the world and converse about it intelligently.

This project directly addresses the need for an AI-powered chatbot that can analyze user-uploaded images, recognize the objects and context within them, and engage in a meaningful, natural language dialogue about the visual content.

2. Proposed Solution

We are developing an interactive, web-based Conversational Image Recognition Chatbot. This application will provide users with a seamless interface to upload any image and ask questions about it in plain English. The AI will process both the image and the question to generate relevant, accurate, and grammatically correct answers in real-time.

The core of our solution is a sophisticated deep learning model that integrates state-of-the-art computer vision and natural language understanding, specifically a **Visual Question Answering (VQA)** model.

3. Core Technology & Architecture

The project is built on a modern, robust technology stack designed for performance and scalability.

- **AI Model:** We are leveraging a powerful, pre-trained multimodal transformer model, **Salesforce's BLIP (Bootstrapping Language-Image Pre-training)**. This model architecture is designed to understand the deep semantic connections between visual and textual data, making it ideal for VQA tasks.
- **Dataset for Fine-Tuning:** The model's knowledge base can be further enhanced by fine-tuning on large-scale, annotated datasets like **COCO 2017**, which contains hundreds of thousands of diverse images.

- **Application Architecture:** The system is designed as a simple and efficient web application:
 - **Frontend/UI:** A clean, user-friendly web interface built with **Gradio**. This allows for rapid development and provides an intuitive experience for image uploads and interactive chat.
 - **Backend & AI Core:** A Python-based server that loads and hosts the fine-tuned VQA model. It manages user requests, processes the inputs, and serves the AI-generated answers back to the user.

4. Key Features & Functionalities

- **Intuitive Image Upload:** Users can easily upload images (.jpg, .png, etc.) directly from their devices.
- **Natural Language Queries:** Users can ask complex, open-ended questions in natural language.
- **Contextual Understanding:** The AI can identify objects, attributes (e.g., colors, shapes), and spatial relationships within the image.
- **Real-Time Interaction:** The chatbot provides answers instantly, creating a fluid and engaging conversational experience.
- **Cross-Platform Accessibility:** As a web application, the chatbot is accessible from any device with a web browser.

5. Current Status & Next Steps

The project has successfully moved beyond the ideation and implementation phases and is currently at the **Prototype** stage.

- **Current Status:** A complete, end-to-end functional prototype has been built. We have successfully fine-tuned a BLIP model on a subset of the COCO dataset and integrated it into a Gradio web application. The application can be launched, and users can interact with the AI.
- **Identified Challenge:** During testing, we observed that the model fine-tuned on a small data subset (~5,000 images) produced some irrelevant answers. This indicates the model is functional but "under-trained."
- **Immediate Next Step (Strategic Enhancement):** To ensure a high-quality and impressive demonstration for SIH 2025, we will update the prototype to use a pre-trained, expert VQA model ([Salesforce/blip-vqa-base](#)). This model has been trained on massive datasets and provides state-of-the-art accuracy out-of-the-box, guaranteeing a robust user experience.
- **Future Work:** Post-competition, the architecture allows for further fine-tuning on larger or more specialized datasets to cater to specific commercial applications (e.g., medical imaging, e-commerce).

This strategic approach ensures we can deliver a highly polished and intelligent prototype that fully realizes the project's vision within the hackathon's timeframe.